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Midterm II Part A
Electromagnetic Theory
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Magnetostatics
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Q1: If $\boldsymbol{B} = 12x\,\widehat{\boldsymbol{x}} + 25y\,\widehat{\boldsymbol{y}} + cz\,\widehat{\boldsymbol{z}}$, find c.

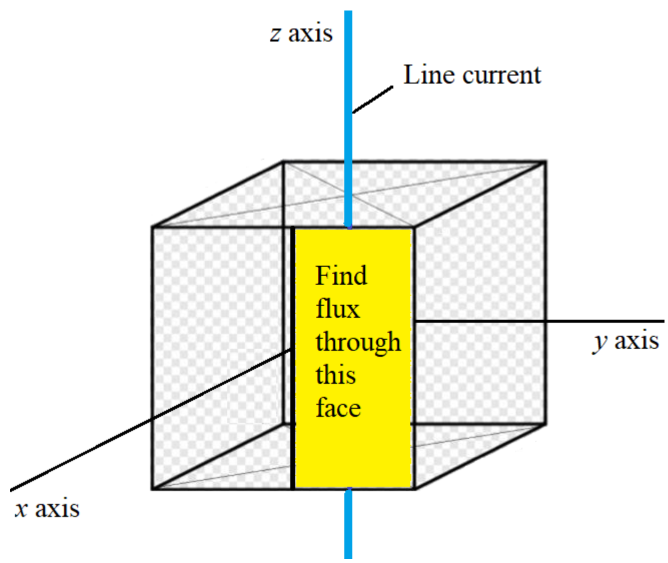
Q2: A cube of edge $2b$ is centered at the origin. A very long, straight wire located along the z axis carries a current I in the z direction. The magnetic flux density is given by

$$\boldsymbol{B} = \frac{\mu_0 I}{2\pi r}\,\widehat{\boldsymbol{\phi}}$$

Show that the flux passing through the surface at $x = b$ bounded by $0 \le y \le b$ and $-b \le z \le b$ is given by

$$\phi = \int_S \boldsymbol{B}\cdot d\boldsymbol{S} = -\frac{\mu_0\,\ln(2)\,b\,I}{2\pi}$$

The required face is shown in the figure below:



With the aid of a sketch explain why there is a negative sign in ϕ.

Q3: Suppose, at a very distant point, the magnetic vector potential is given as

$$\boldsymbol{A} = \frac{\mu_0 I k}{r^2}\widehat{\boldsymbol{z}}$$

where k is some positive constant and I is the current. Find $\boldsymbol{B}$.

Q4: Given that $\boldsymbol{H}_1 = 3\widehat{\boldsymbol{x}} - 2\widehat{\boldsymbol{y}} + 2\widehat{\boldsymbol{z}}$ A/m in region $2x - y - 1 \leq 0$, where $\mu_1 = 4\mu_0$, calculate (a) $\boldsymbol{M}_1$ and $\boldsymbol{B}_1$ (b) $\boldsymbol{H}_2$ and $\boldsymbol{B}_2$ in region $2x - y - 1 \geq 0$, where $\mu_2 = 3\mu_0$.

Formulae

$$\int \frac{u}{u^2 + a^2} du = \frac{1}{2} \ln\left(a^2 + u^2\right) + \text{ constant}$$

Cylindrical to Rectangular

$$\text{Variable change} \begin{cases} r = \sqrt{x^2 + y^2} \\ \phi = \tan^{-1}\left(\dfrac{y}{x}\right) \\ z = z \end{cases} \qquad \begin{cases} \sin\phi = \dfrac{y}{\sqrt{x^2 + y^2}} \\ \cos\phi = \dfrac{x}{\sqrt{x^2 + y^2}} \end{cases}$$

The relationships between $(\widehat{\boldsymbol{x}}, \widehat{\boldsymbol{y}})$ and $(\widehat{\boldsymbol{r}}, \widehat{\boldsymbol{\phi}})$:

$$\widehat{\boldsymbol{x}} = \cos\phi\,\widehat{\boldsymbol{r}} - \sin\phi\,\widehat{\boldsymbol{\phi}}$$
$$\widehat{\boldsymbol{y}} = \sin\phi\,\widehat{\boldsymbol{r}} + \cos\phi\,\widehat{\boldsymbol{\phi}}$$

or

$$\widehat{\boldsymbol{r}} = \cos\phi\,\widehat{\boldsymbol{x}} + \sin\phi\,\widehat{\boldsymbol{y}}$$
$$\widehat{\boldsymbol{\phi}} = -\sin\phi\,\widehat{\boldsymbol{x}} + \cos\phi\,\widehat{\boldsymbol{y}}$$